CO2xN: Porewater Methane

Month Year Samples

2025-08-22

Contents

1 Read in data files **2**

2 Assess standard curves **2**

3 Aanlayze the Check Standards **5**

4 Analyze the Blanks **6**

5 Dilution correct samples and Check Range **9**

6 Calculate gas in water **10**

7 Merge samples with Metadata & check all samples are present **10**

8 Visualize Data **10**

##Run Information

```r
cat("Run Information: Brenden D. Blakley ") #lets you know what section you're in
```

```
## Run Information: Brenden D. Blakley
```

```r
#set the run date & user name
run_date <- "06/12/2025"
sample_year <- "2025"
sample_month <- "May"
user <- "Brenden Blakley"
#identify the files you want to read in
#read in as a list to accommodate multiple runs in a month

GHG_file <- "Raw Data/20250612_2M_MAY_GHG.csv"


# Define the file path for QAQC log file - NO Need to change just check year
log_path <- "Processed Data/LTREB_2M_ShimadzuGC_QAQC_2025.csv"

# Define final path for data be sure to change year and month and run number
final_path <- "Processed Data/GCReW_CO2xN_Porewater_CH4_202505_1.csv"

#Gas in Water Calculations
# "TempC" column value in Celsius (equilibration temperature) and volume of gas (in liters) in syringe
TempC = 22.7 #Equilibration Temperature (Celsius)
volume = 0.012 #volume of gas (in liters) in syringe

#record any notes about the run or anything other info here:
run_notes <- ""

#Set up file path for metadata
#downloaded metadata csv - downloaded from Google drive as csv for this year
Raw_Metadata = "Raw Data/Sample_IDs_LTREB_SMARTX_2024.csv"
#Collection_Dates = "Raw Data/Sample_Collection_Dates_2024.csv"

cat(run_notes)
```

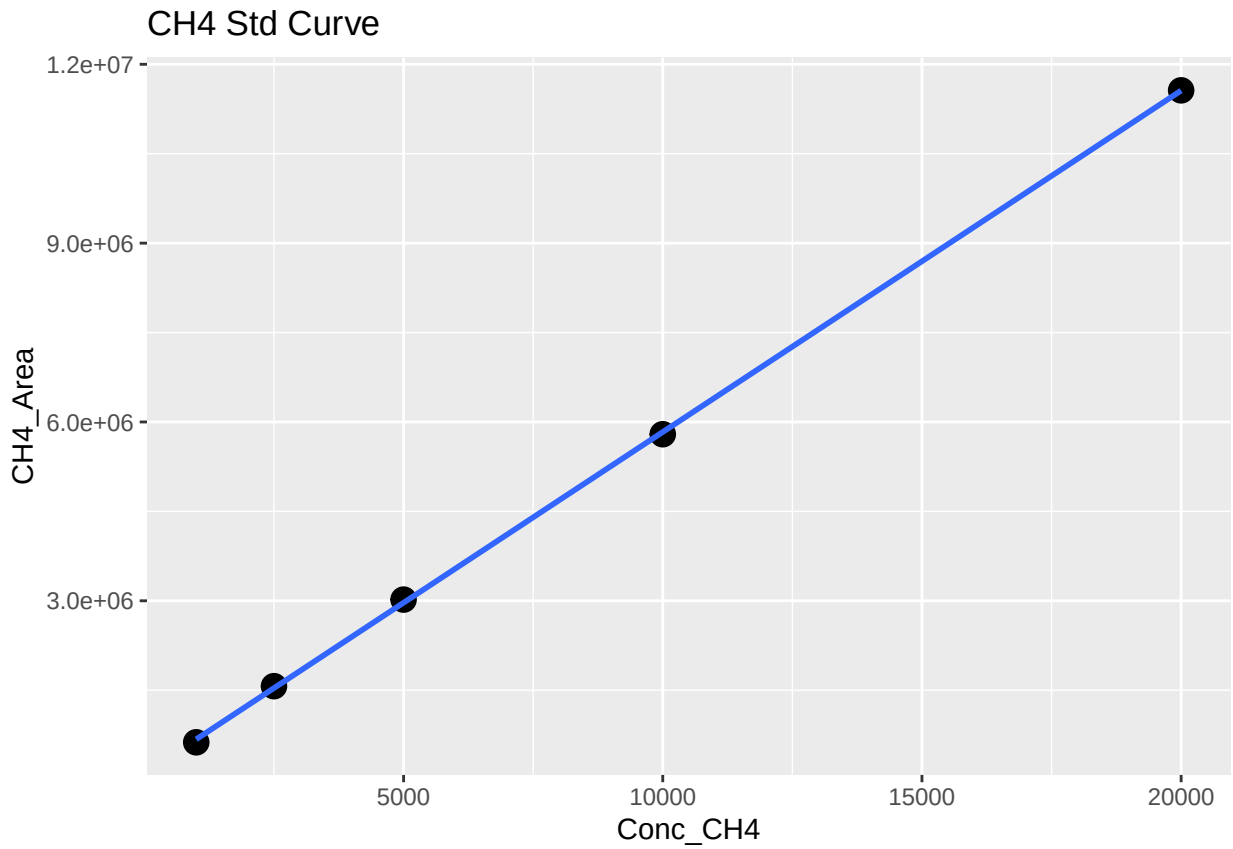
##Set Up

##Read in metadata and create similar sample IDs for matching to samples

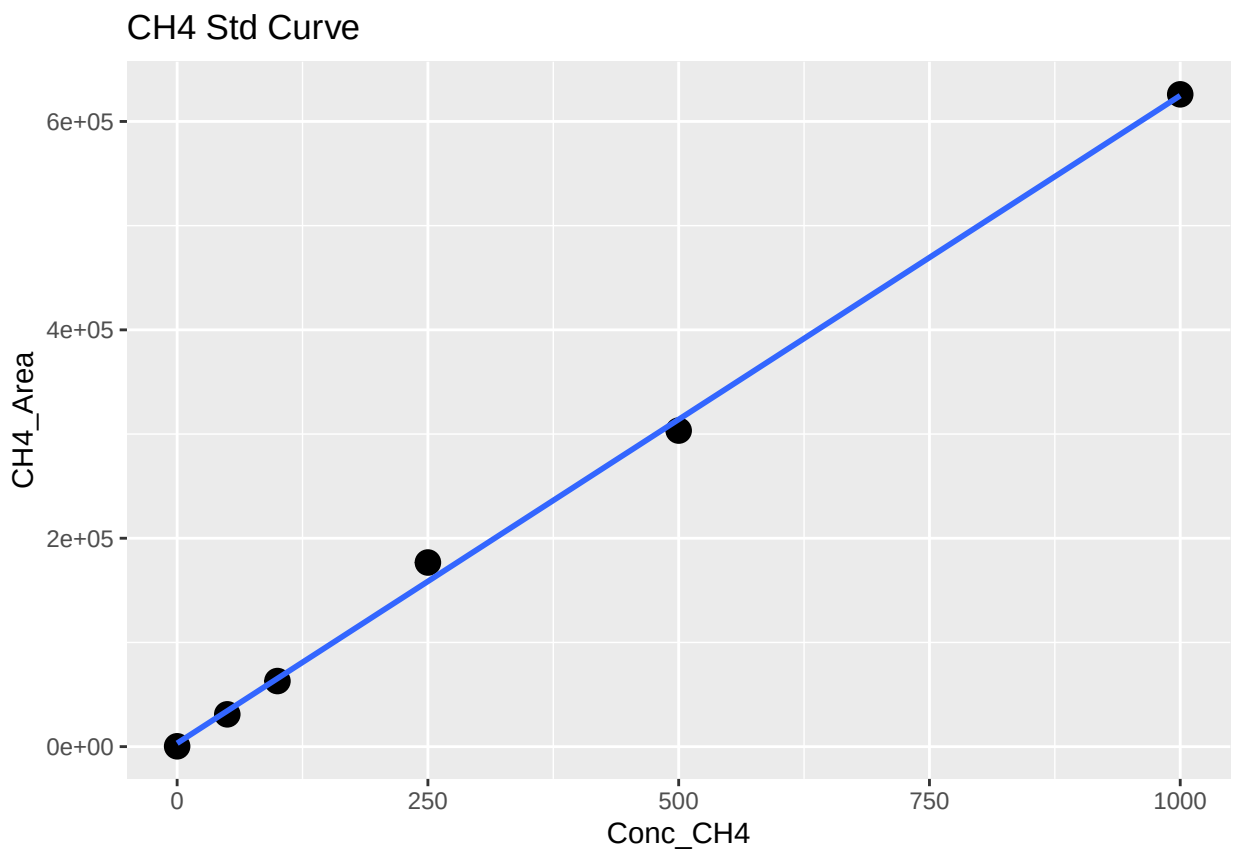
1 Read in data files

2 Assess standard curves

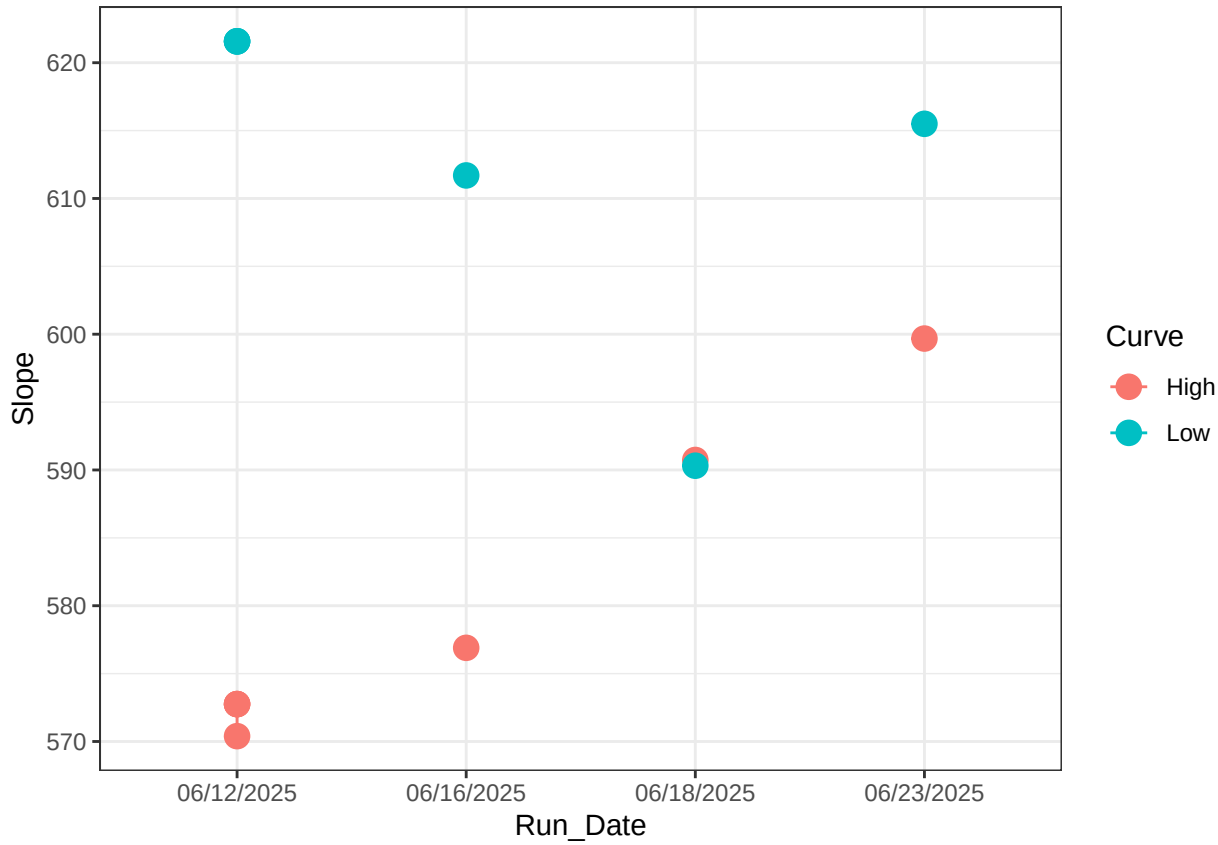
```
## `geom_smooth()` using formula = 'y ~ x'
```




```
## `geom_smooth()` using formula = 'y ~ x'
```




```
##   X Curve        R2    Slope    Intercept   Run_Date
## 1 1  High 0.9986295 576.8990  173986.2174 06/16/2025
## 2 2   Low 0.9990260 611.6961   -3791.2684 06/16/2025
## 3 3  High 0.9999101 570.3925  138747.3478 06/12/2025
## 4 4   Low 0.9979098 621.5734    3224.2535 06/12/2025
## 5 5  High 0.9989882 590.7374 -109278.7391 06/18/2025
## 6 6   Low 0.9990995 590.3138     954.7682 06/18/2025
```



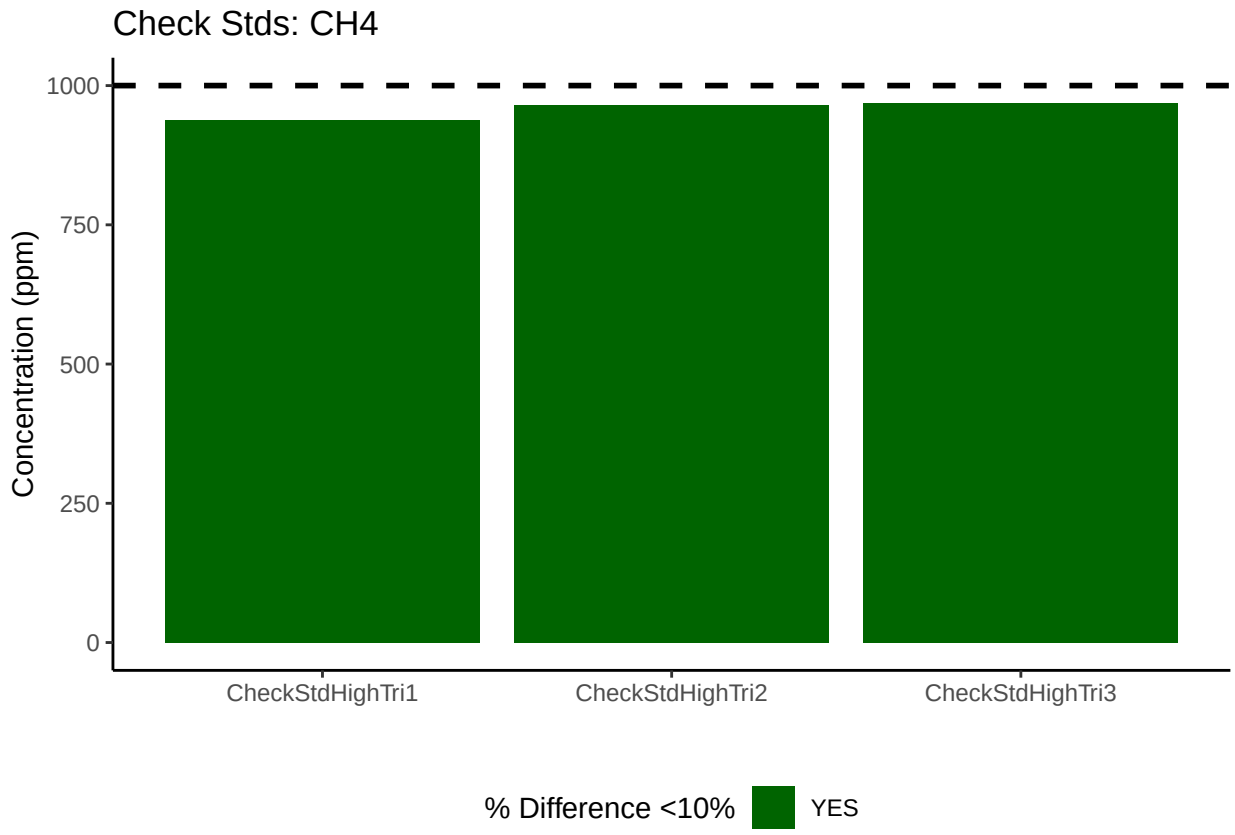

```
## High_R2 Good
## Low_R2 Good
```

2.1 Now calculate the CH4 & CO2 concentrations in ppm

3 Aanlayze the Check Standards

```
## >60% of CH4 Check Standards are within range of expected concentration - PROCEED
```

```
## Warning: Using ‘size‘ aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use ‘linewidth‘ instead.
## This warning is displayed once every 8 hours.
## Call ‘lifecycle::last_lifecycle_warnings()‘ to see where this warning was
## generated.
```



4 Analyze the Blanks

```
## Assess Blanks
```

```
## >60% of CH4 Blanks are below the lower 25% quartile of samples - PROCEED
```

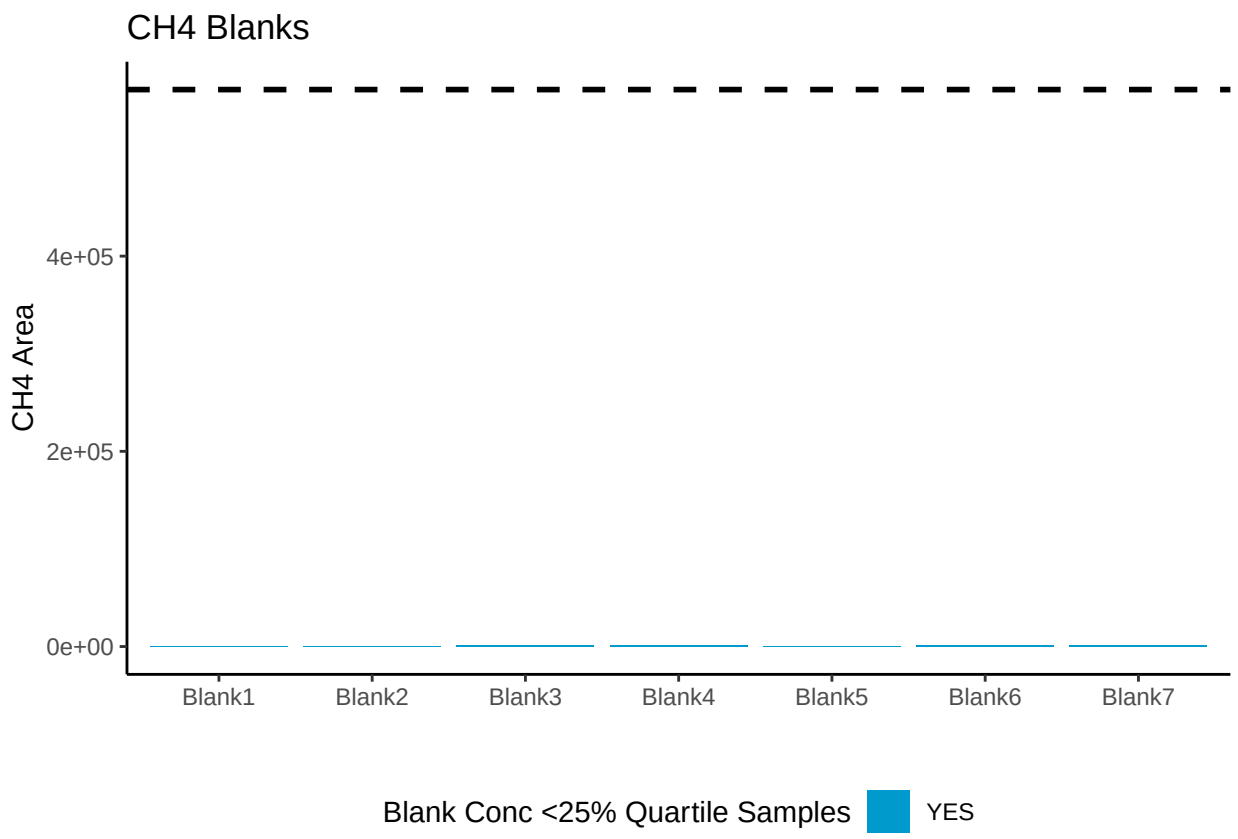
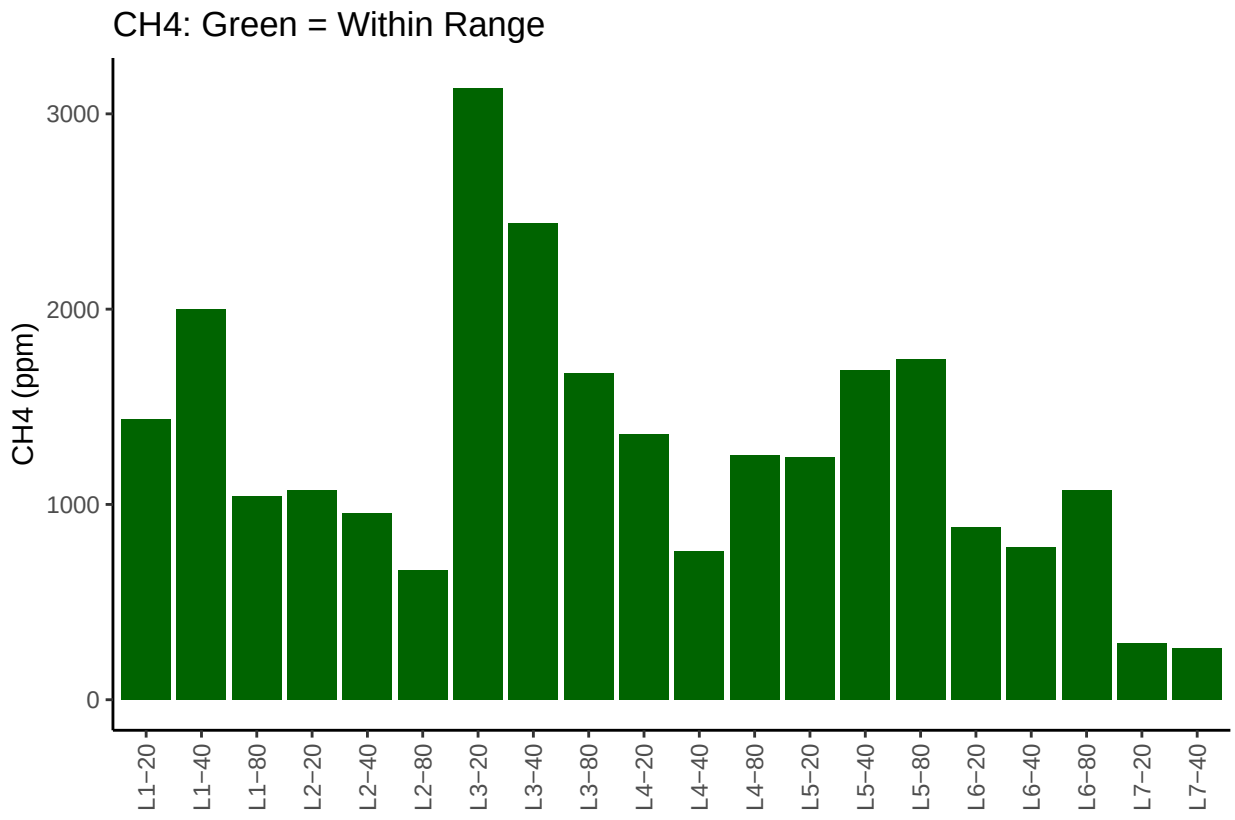
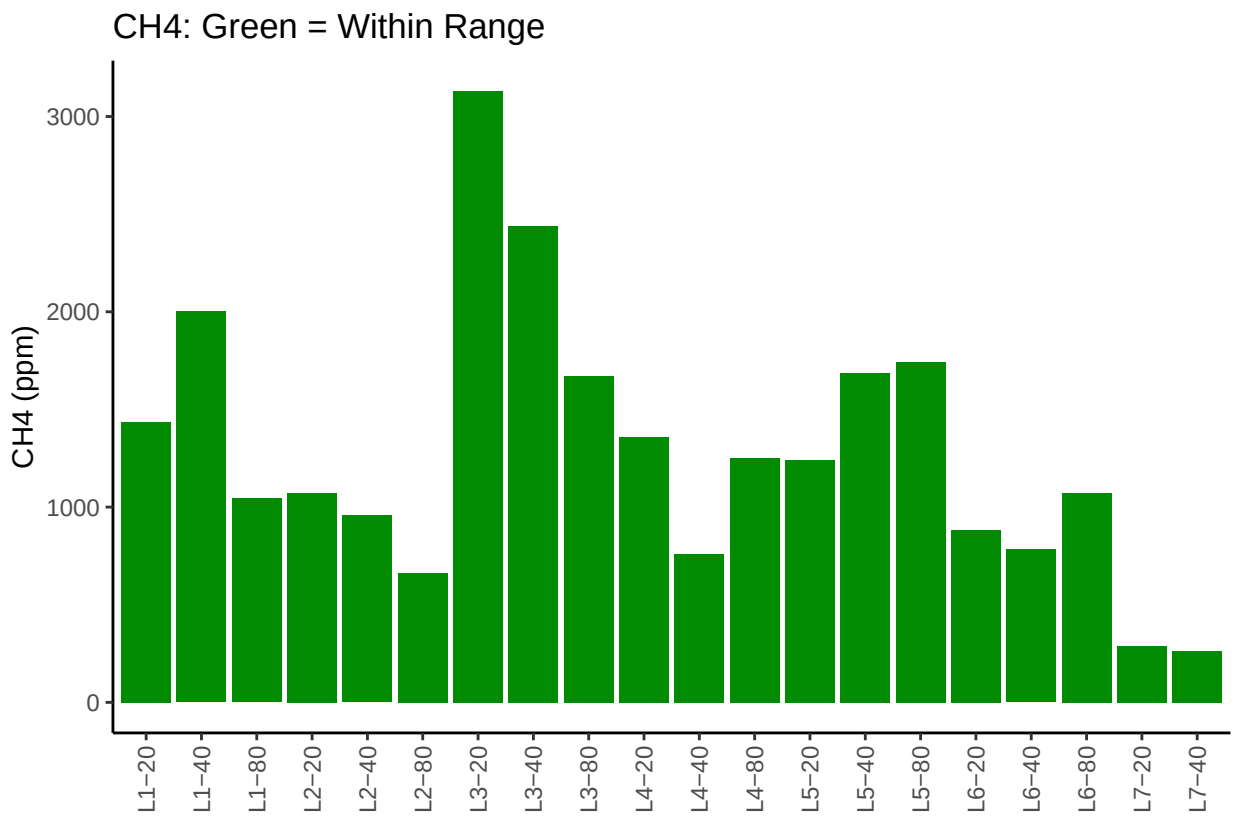


Table 1: Mean Area of Blanks

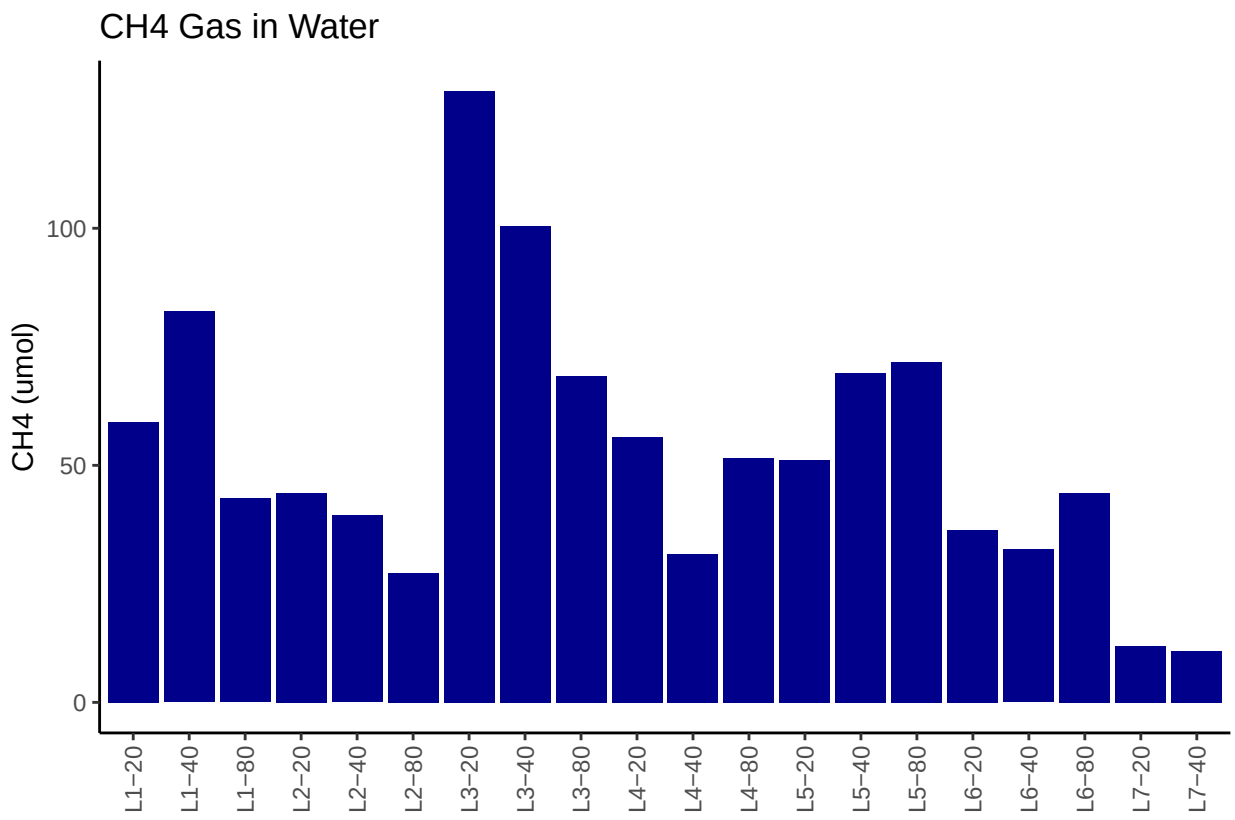
Test	Blank_Mean_Area
CH4	1294.143

5 Dilution correct samples and Check Range





6 Calculate gas in water



7 Merge samples with Metadata & check all samples are present

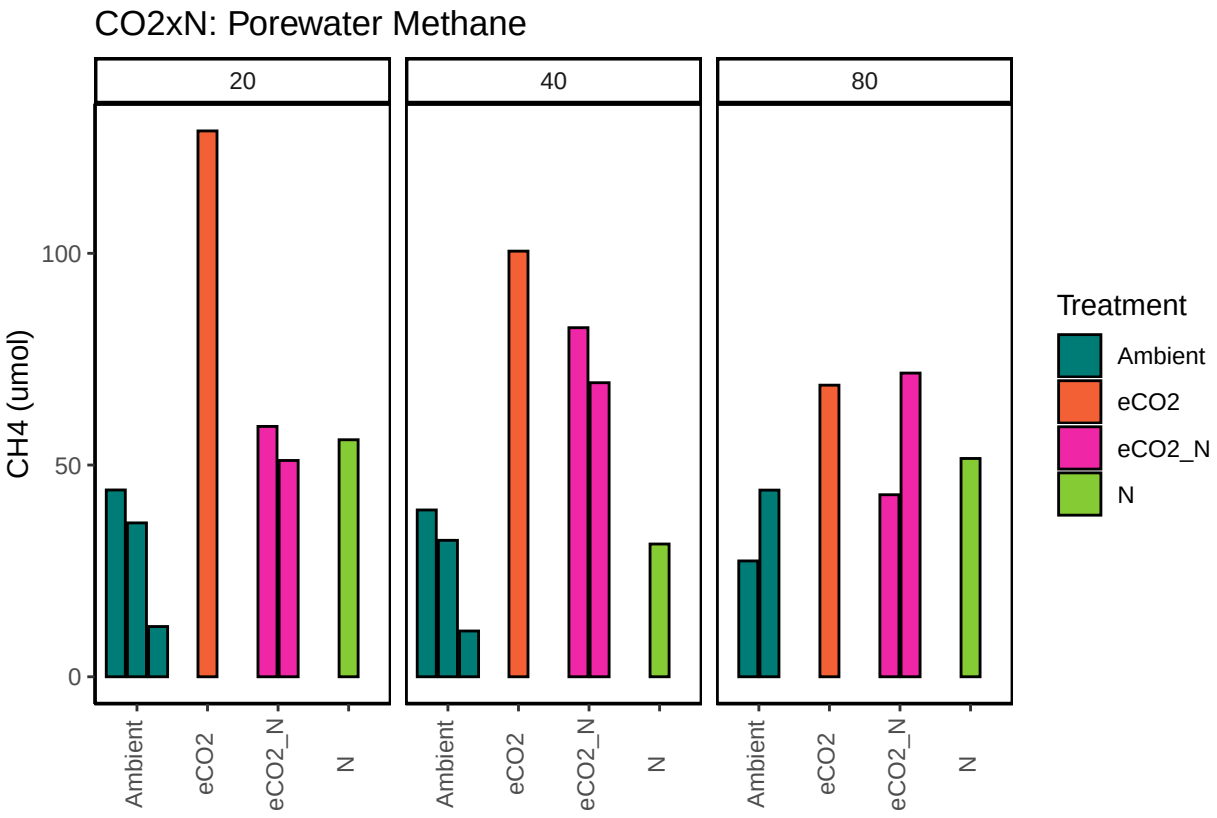
```
## Merge Samples with Metadata
```

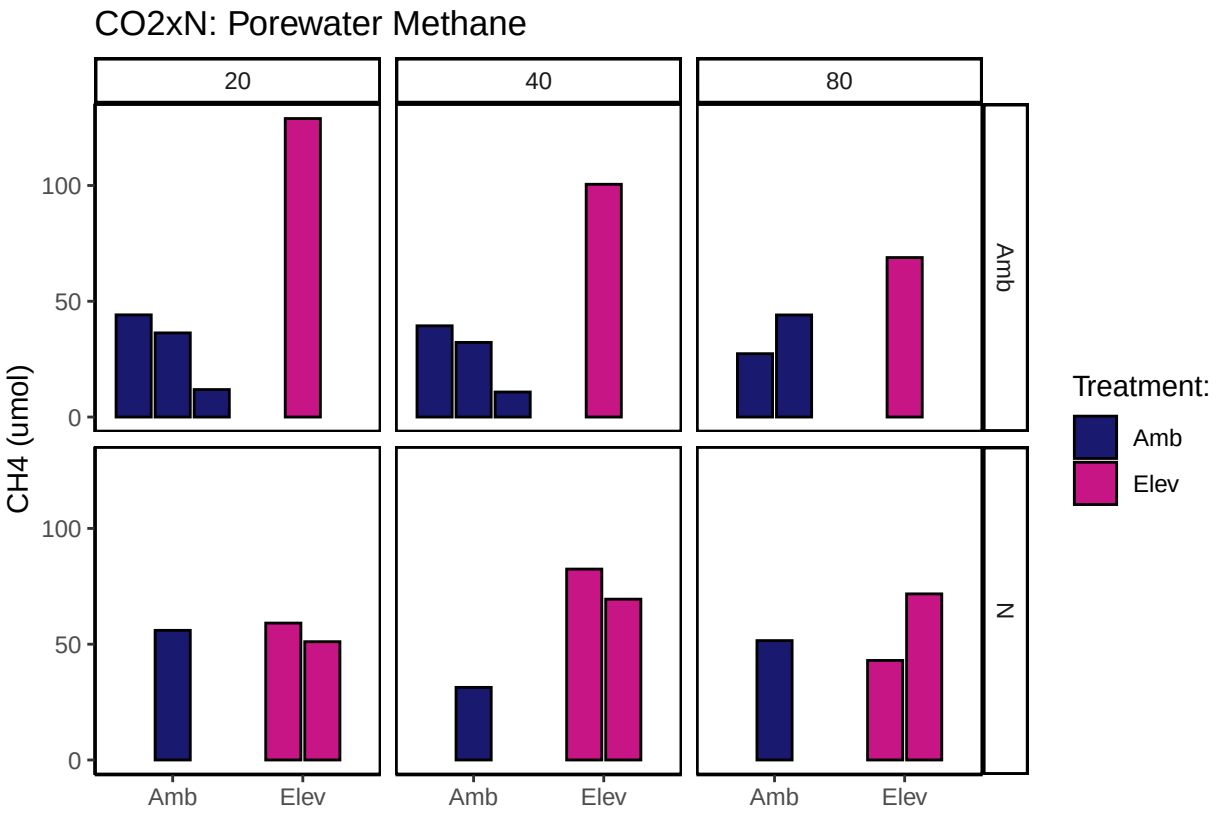
```
## Some sample IDs are missing from data.
```

```
##  [1] "L7-80"  "L8-20"  "L8-40"  "L8-80"  "L9-20"  "L9-40"  "L9-80"  "L10-20"
##  [9] "L10-40" "L10-80" "L11-20" "L11-40" "L11-80" "L12-20" "L12-40" "L12-80"
## [17] "L13-20" "L13-40" "L13-80" "L14-20" "L14-40" "L14-80" "L15-20" "L15-40"
## [25] "L15-80" "L16-20" "L16-40" "L16-80" "L17-20" "L17-40" "L17-80" "L18-20"
## [33] "L18-40" "L18-80" "L19-20" "L19-40" "L19-80" "L20-20" "L20-40" "L20-80"
```

8 Visualize Data

```
## Visualize Data
```


CO2xN: Porewater Methane
20
40
80
CH4 (umol)
100
50
0
Ambient
eCO2
eCO2_N
N
Treatment
Ambient
eCO2
eCO2_N
N



8.1 Export Processed Data

```
## [1] "06/12/2025"
```

#end